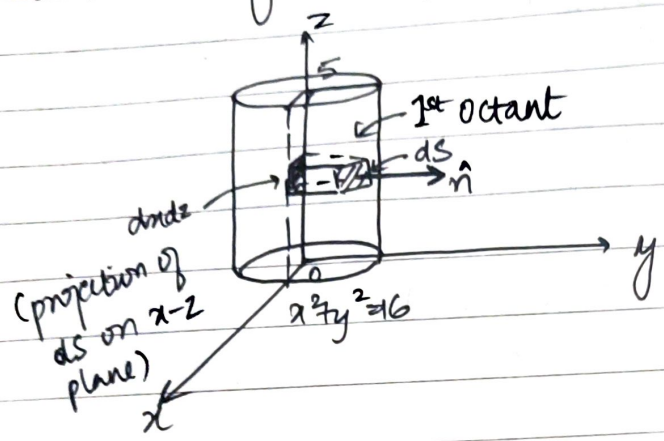
Tut 2 Physics

Q1 $\iint A.\hat{n}\,ds$

$A = z\hat{i} + x\hat{j} - 3y^2$

The region of integration is the shaded region (3) which is curved surface of the cylinder $x^2+y^2=16$ $(z=0 \text{ to } z=5)$



~~The region of int^n is~~

Consider an area element ds on curved surface S having unit normal ~~at~~ $\hat{n}$ as shown in fig. Project the area element ds on the x-z plane and this element is dxdz in the region R

$$\iint_S \bar{A}.\hat{n}\,ds = \iint_R \bar{A}.\hat{n}\,\frac{dxdz}{|\hat{n}.\hat{j}|} \qquad \left(\because ds = \frac{dxdz}{|\hat{n}.\hat{j}|}\right)$$

The normal is given by

$$\vec{n} = \vec{\nabla}\phi \qquad (\phi = c) \quad (x^2+y^2=16)$$

$$\vec{n} = \vec{\nabla}(x^2+y^2)$$
$$= 2x\hat{i} + 2y\hat{j}$$

$$\hat{n} = \frac{2x\hat{i}+2y\hat{j}}{2\sqrt{x^2+y^2}} = \frac{2x\hat{i}+2y\hat{j}}{2\sqrt{16}}$$
$$= \frac{x}{4}\hat{i} + \frac{y}{4}\hat{j}$$

$\hat{n}\cdot\hat{j} = \frac{y}{4}$

$\vec{A}\cdot\hat{n} = (z\hat{i} + x\hat{j} - 3y^2z\hat{k})\left(\frac{x}{4}\hat{i} + \frac{y}{4}\hat{j}\right)$

$= \frac{xz}{4} + \frac{xy}{4}$

The surface integral is

$$\iint_R \left(\frac{xz + xy}{4}\right) \frac{dx\,dz}{y/4}$$

$$= \int_0^5 \int_0^4 \left(\frac{xz}{\sqrt{16-x^2}} + x\right) dx\,dz$$

$= I_1 + I_2$

$$I_1 = \int_0^5 z\,dz \int_0^4 \frac{x}{\sqrt{16-x^2}}\,dx$$

$$= \frac{25}{2}\left[-\sqrt{16-x^2}\right]_0^4$$

$16 - x^2 = t$

$-2x\,dx = dt$

$x\,dx = \frac{-dt}{2}$

$\frac{-1}{2}\int \frac{dt}{\sqrt{t}} = \frac{-1}{2}\left[\left(\frac{-2}{1}\right) t^{+1/2}\right]$

$$= \frac{+25}{2}\left[-\sqrt{16-16} + 4\right]$$

$= \boxed{50}$

Q2 $\vec{F} = ax\hat{i} - by\hat{j}$

(a) $W = \int F\cdot dr = \int F_x dx + \int F_y dy = -\int ax\,dx - \int by\,dy$

Along path PO ,

$y \to 0$ & $x = 1$ to $x = 0$

$$\therefore W = -\int_1^0 ax\,dx = \frac{a}{2}$$

<u>Along OQ</u>

$x = 0$, $y = 0 \to 1$

$$W = -\int_0^1 by\,dy = \frac{-b}{2}$$

$$W(PO + OQ) = \frac{a}{2} - \frac{b}{2} = \int_{PO \to OQ} \bar{F} \cdot d\bar{r}$$

- Along curved path PQ, the eq^n of curved surface PQ is $x^2 + y^2 = 1$.

$$W = \int F_x\,dx + \int F_y\,dy$$

$$y = \sqrt{1 - x^2}$$

$$dy = \frac{-x}{\sqrt{1-x^2}}\,dx$$

As x inc., y dec. along the path QP.

$$W = \int_1^0 -ax\,dx + \int_1^0 b\left(\cancel{\sqrt{1-x^2}}\right)\frac{x}{\cancel{\sqrt{1-x^2}}}\,dx$$

$$= \frac{a}{2} - \frac{b}{2} \quad \text{along PQ}$$

(b) Work done along 2 paths are same. This is an indication of conservative force.

If a line integral of a vector func^n is same along diff. paths connecting same 2 points then the vector func^n is conservative

(c) A conservative vector func^n can be expressed as the gradient of a scalar func^n ϕ

$$\vec{F} = \vec{\nabla}\phi$$

$$\Rightarrow -ax\hat{i} - by\hat{j} = \frac{\partial}{\partial x}\phi\,\hat{i} + \frac{\partial}{\partial y}\phi\,\hat{j}$$

$$\frac{\partial \phi}{\partial x} = -ax \qquad \frac{\partial \phi}{\partial y} = -by$$

$$\phi(x,y) = \frac{-ax^2}{2} + f(y) \rightarrow (1)$$

$$\phi(x,y) = \frac{-by^2}{2} + g(x) \rightarrow (2)$$

Comparing (1) & (2),

$$f(y) = -\frac{by^2}{2}$$

$$g(x) = \frac{-ax^2}{2}$$

$$\phi(x,y) = \frac{-ax^2}{2} - \frac{by^2}{2}$$

P.E.(U) can be written as

$$F = -\frac{dU}{dx}$$

$$U = -\phi = \frac{ax^2}{2} + \frac{by^2}{2}$$

This is the P.E. of 2D harmonic oscillator & $\vec{F}$ is a 2D spring force.

(d) The curl of any conservative force is always zero.

$$\vec{\nabla}\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -ax & -by & 0 \end{vmatrix}$$

$$= (0-0)\hat{i} + (0-0)\hat{j} + (0-0)\hat{k}$$

$$= \underline{\underline{0}}$$

This is consistent with the results (a), (b) & (c)

We know $\overset{\text{curl}}{\vec{\nabla}}\times(\overset{\text{grad}}{\vec{\nabla}}\phi) = 0$ for any scalar func^n ϕ. Hence $\vec{F}$ must be equal to $\vec{\nabla}\phi$

Q3 (a) $\nabla^2 g(x,y,z)$

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\left(\frac{1}{12}x^4 - \ln y + \ln z\right)$$

$$= \frac{\partial}{\partial x}\left(\frac{\cancel{4}x^3}{\cancel{12}3}\right) + \frac{\partial}{\partial y}\left(-\frac{1}{y}\right) + \frac{\partial}{\partial z}\left(\frac{1}{z}\right)$$

$$= x^2 + \frac{1}{y^2} - \frac{1}{z^2}$$

(b) $\nabla^2\vec{V}(x,y,z) = e^{-xyz}\hat{i} + z^2\hat{j} + xz\hat{k}$

Q3) (a) $g(x,y,z) = \frac{1}{12}x^4 - \ln\frac{y}{z}$

$$\nabla^2 g(x,y,z) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\left(\frac{1}{12}x^4 - \ln y + \ln z\right)$$

$$= \left(x^2 + \frac{1}{y^2} - \frac{1}{z^2}\right) \rightarrow A$$

(b) $\vec{v}(x,y,z) = e^{-xy^2}\hat{i} + z^2\hat{j} + xz\hat{k}$

$$\nabla^2 g(x,y,z) = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\left(e^{-xy^2}\hat{i} + z^2\hat{j} + xz\hat{k}\right)$$

$$\cancel{= i\left(e^{-xy^2}(-y^2)\right)}$$

$$= i\left(e^{-xy^2}(y^2)^2\right) + i\left(e^{-xy^2}(xz)^2\right) + 2j + i\left(e^{-xy^2}(xy)^2\right)$$

$$= 2j + \hat{i}\left[e^{-xy^2}\left(y^2z^2 + x^2z^2 + x^2y^2\right)\right] \rightarrow B$$